Heat and Mass Transfer Final Project Report

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Battery Cooling Via Liquid Convection

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Executive Summary

Effective heat management is critical for EV battery safety, performance, and durability (Zamanov, 2024). The project used a lab-scale experiment to assess the performance of liquid coolants in battery cooling systems. A heated aluminum plate simulated a battery module that was cooled by an external flow of temperature-controlled liquid. The objective was to quantify and evaluate the cooling effectiveness of pure water and brine solutions at different concentrations. The experimental setup featured a Cole-Parmer refrigerated circulating bath flowing coolant across an inclined heated slide. Temperatures were measured using thermocouples, and the data was collected using LabVIEW. Tests varied in the coolant type (water and brines at 0.5, 1, and 2 lb/gal), beginning plate temperature (50°C and 80°C), and interface conditions to measure thermal resistance.

The analysis of transient temperature data was carried out using thermocouple data and provided crucial parameters: convective heat transfer coefficient (h), total thermal resistance (R_{th}), and cooling time (t). The heat transfer coefficient for each test was calculated using a relation with area, temperature, and heating element power. Cooling time was determined by analyzing the temperature datasheets to determine when the apparatus reached steady state. Finally, thermal resistance was calculated using a relation with temperature and heating element power. The results led to clear conclusions. Pure water was the most efficient coolant, with the highest h. Brine solutions consistently reduced h, resulting in poorer performance. Cooling was less successful at higher temperatures, with considerably longer cooling periods at 80°C compared to 50°C. Furthermore, thermal resistance contributed significantly to the steady state temperature. The key takeaway is that for direct liquid cooling applications, water outperforms the tested brine solutions. The project demonstrates the importance of fluid properties and interfacial design for thermal management system efficiency. These findings provide practical guidelines for the design of EV battery cooling systems, with coolant selection and interface quality being critical engineering factors.

1. Introduction

Electric vehicles (EVs) are a driving force in the modern automobile industry's transition to sustainability, due to breakthroughs in battery technology and environmental regulations. One major engineering problem in their development is regulating the enormous waste heat produced by high-density lithium-ion batteries during high-power operation. This thermal challenge is most acute during demanding conditions such as fast charging or aggressive acceleration. Without proper thermal management, this heat can significantly reduce battery performance, shorten lifespan, and present serious safety risks, like thermal runaway (Zamanov, 2024).

Liquid cooling is a prevalent solution due to its higher heat capacity and superior thermal conductivity compared to air. Engineered coolants, such as ethylene or propylene glycol, are widely used in industry because of their freeze protection and dielectric qualities. Glycol-based coolants were avoided in this experiment due to high cost and special safety handling requirements. Instead, the project looks at more readily available options: pure water and brine (saltwater) solutions. Comparing water to brines provides a direct examination of how increasing fluid salinity and therefore modifying important parameters such as viscosity and thermal conductivity influences cooling performance. This provides practical, comparative data on the cooling effectiveness of different fluids for a simulated battery thermal management system.

The primary objective is to quantify thermal performance of water against different brine concentrations. Key analytical parameters include the convective heat transfer coefficient, total thermal resistance, and cooling transient time. A secondary objective is to assess the impact of thermal resistance at assembly interfaces, a practical element that is frequently simplified in theoretical models.

To accomplish this, a laboratory scale system was built in which a heated aluminum plate serves as a simpler battery cell and is cooled by a temperature-controlled flow of liquid. This report describes the experimental technique, provides a comparative analysis of the findings, and examines the practical implications for thermal system design.

2. Experimental Method

The goal of the experiment is to cool a heater using liquid convection, changing one variable for each test, to compare the effectiveness of liquid convection under various conditions. To achieve this goal, the experiment is designed so that water of a constant temperature flows over the heater evenly and at a constant rate. A slide is used to achieve this. The slide is made of bent aluminum sheet metal, and the heating pad (a 4x4 in square) is taped under it. To insulate the heater, Styrofoam is attached underneath the heater and on the sides of the aluminum slide, so that the heat only leaves through the surface where water will flow. The slide is 4 in wide on the bottom, with sides that are 2 in. tall. The slide is 2 ft long and has a thickness of 0.025 in.

When the experiment is being run, water from a tube falls onto a slide and spreads evenly over the plate before running over the heated portion of the slide. The water cools this portion via convection before falling off the far end of the slide. There are six thermocouples taking temperature readings: one in the tube measures incoming water temperature, one at the end measures exiting water temperature, one in the basin measures ambient water temperature, and three on the heater measure the heater temperatures. **Figure 2.1** shows schematics of the convective slide with thermocouple locations. **Figure 2.2** shows an image of the experimental setup in action, with water running down the slide. To support the slide and contain the liquid, two tote bins are used. Rope secures the slide to one tote bin.

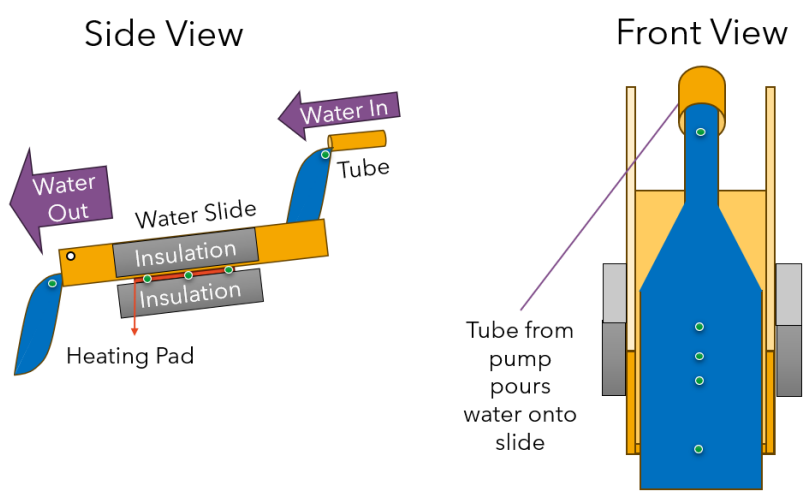

Figure 2.1 Schematics of the side view and front view of the convective slide. Green dots represent thermocouple locations

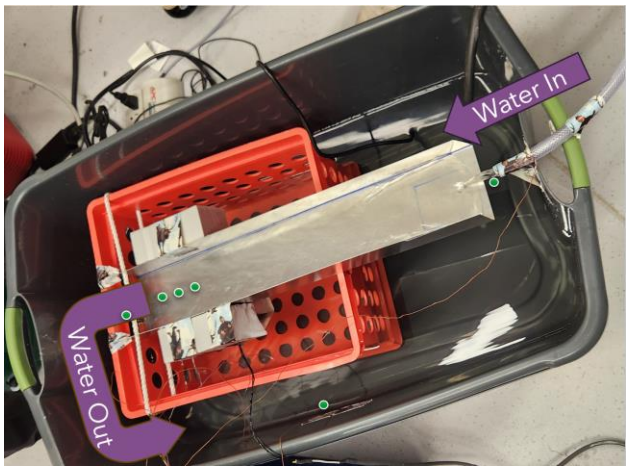

Figure 2.2 Image of the experimental setup with the convective slide. Green dots represent thermocouple locations

When water falls off the slide, it falls into a basin. A small pump pumps water from the basin to the Cole-Parmer Refrigerated Circulating Bath (Serial # 1911-02821, referred to as the CP pump from here on out). The advantage of the CP pump is that it keeps the liquid it pumps at a constant temperature by storing it inside of itself and applying heating or cooling. This ensures that the water being pumped over the slide remains at a constant temperature. Water from the CP pump is pumped to the slide at a relatively constant flow rate of 0.84±0.02 gallons per minute (gpm), where it falls onto the slide, completing the loop. **Figure 2.3** shows the schematic of the water flow in the experimental setup. Figures 2.4a and 2.5b show images of the full experimental setup.

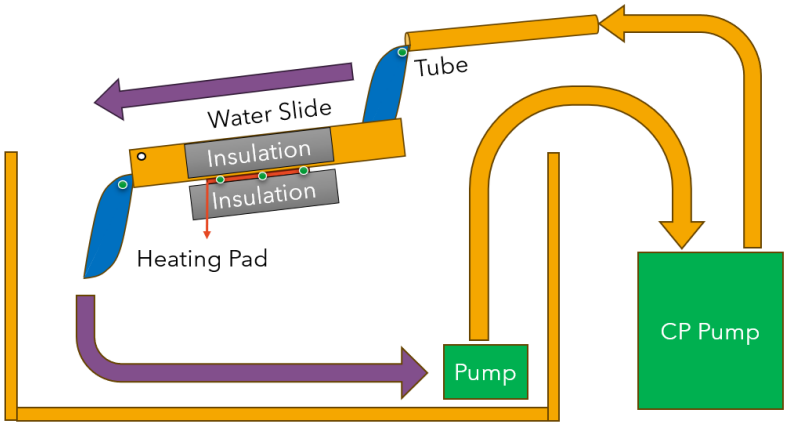

Figure 2.3 Schematic of the water flow in the experimental setup. Green dots represent thermocouple locations

Figure 2.4a (left) and **2.4b** (right) Images of the experimental setup. **Figure 2.4a** shows the full experimental setup, and **Figure 2.4b** shows a top view of the smaller pump.

To collect thermocouple data, a laptop from the electronics shop was used to run LabView software.

Several improvements were made to the design. To drive water flow, a rain barrel and gravity were initially going to be used. The team decided to use a pump so avoid the need to manually move the water back into the barrel, making for less work and less liquid needed in the experiments. The CP pump was also added to the setup for its cooling abilities. Also, to evenly distribute water from the hose onto the slide, the team was initially going to make an attachment for the hose to spread the water, but after playing with the setup, the team realized that, at the pump's flow rate, as long as the slide had enough length before the heated section, the water spread evenly before the heated section.

The testing methodology is as follows:

1. Set up the experiment.
2. Plug the heating pad into the powerstat and set the voltage.
3. Wait for steady state temperature to be reached.
4. Start recording temperature data from thermocouples using LabView software.
5. Turn on pumps. Water should flow over plate.

6. Wait for steady state temperature to be reached.

7. End data acquisition.

8. Repeat tests for each variable

Three variables were tested. The voltage to the plate was varied to 32 V or 53 V such that the pad temperatures were initially 50°C or 80°C. Salt concentration was varied to 0, half, 1 and 2 pounds per gallon. Thermal and contact resistance was tested by adding an extra aluminum plate of thickness 0.050 in between the slide and heat pad, as shown in **Figure 2.5**.

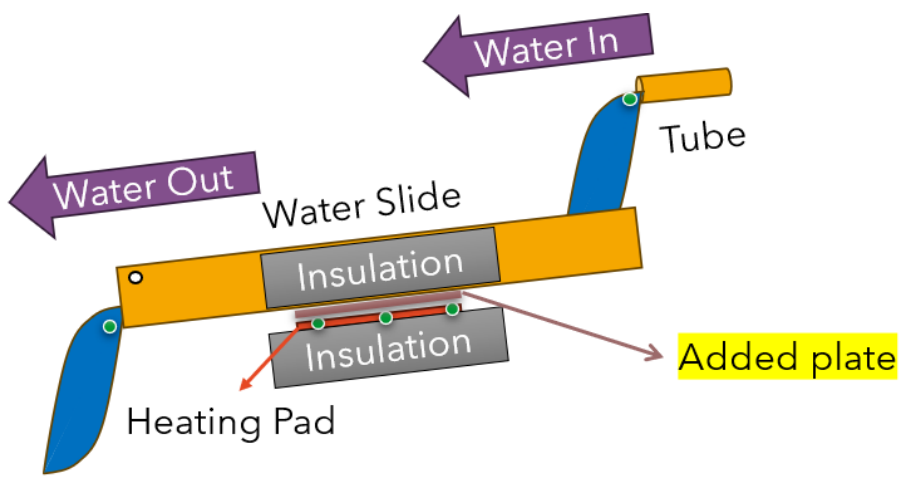

Figure 2.5 Schematic of the convective slide with the added plate. Green dots represent thermocouple locations

In addition, **Table 2.1** converts the values in lb/gal to their corresponding salt concentrations in percentage form for clearer reference and intuition.

Table 2.1 lb/gal converted to Salt concentration **percentage**

lb/gal	**Salt Conc. In Percentage**
½ lb/gal	5.7%
1 lb/gal	10.7%
2 lb/gal	19.4%

Ten tests were run in total; **Table 2.2** is a matrix showing all the tests run and what variables were changed each run.

Table 2.2 Test matrix with all runs and variable settings each run

Test Number	*Initial Heat Pad Temperature*	*Salt Concentration*	*Plate added?*
1	50°C	0	No
2	80°C	0	No
3	50°C	0	Yes
4	80°C	0	Yes
5	50°C	½ lb/gallon	No
6	80°C	½ lb/gallon	No
7	50°C	1 lb/gallon	No
8	80°C	1 lb/gallon	No
9	50°C	2 lb/gallon	No
10	80°C	2 lb/gallon	No

3. Analysis

The analysis for the experiment utilizes the thermocouple transient temperature data to provide values for the relevant parameters; convective heat transfer coefficient (h), total thermal resistance (R_{th}), and cooling time (t). The heat transfer coefficient and cooling time were determined for all test cases, while thermal resistance was only determined for the test cases with pure water. The analysis was carried out using several assumptions; heat transfer was one-dimensional, the apparatus had perfectly insulated sides, radiative heat transfer had a negligible effect on the system, and there was a negligible temperature rise within the coolant. The thermocouples located in the coolant flow before and after the heated section were utilized to validate the assumption that there was a negligible temperature rise within the fluid.

Temperature data from each of these thermocouples was analyzed, and it was determined that the change in measured temperature and the temperature difference between them was less than 0.25 °C. This value can be considered negligible in the context of this experiment.

The convective heat transfer coefficient was calculated using an energy balance for the control volume depicted in **Figure 3.1**, which was used to determine **Equation 3.1**. This method states that all energy entering the system comes from the heating element under the slide, and all energy leaving the system is transferred out by the water flowing over the slide. The full derivation for **Equation 3.1** from the control volume energy balance is located in **Appendix A**, along with a sample calculation.

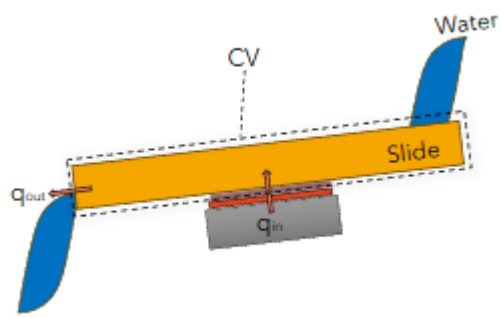

Figure 3.1: Control Volume for Experimental System

$$h = \frac{q}{A(T_s - T\infty)}$$

Equation 3.1: Convective Heat Transfer Coefficient (h)

Thermal resistance of the added aluminum plate was determined by first calculating the total thermal resistance of the system with and without the plate using **Equation 3.2**. The diagram in **Figure 3.2** shows the difference in the thermal resistance circuit of the system in both configurations. It is important to note that this simplified diagram does not explicitly mention the contact resistance between each surface. Instead, it is modeled that the contact resistance is a portion of the resistance of each component in the system.

The added aluminum plate will add both contact and conduction thermal resistance. The diagram shows that the only change to the total system thermal resistance is the resistance of the plate.

Therefore, the plate's thermal resistance can be found by using **Equation 3.3** to take the difference between the two configurations' total thermal resistance. The full derivation for **Equation 3.2** and **Equation 3.3** can be found in **Appendix B**, along with a sample calculation.

$$R_{total} = \frac{T_s - T_\infty}{q}$$

Equation 3.2: Total Thermal Resistance (R_{th})

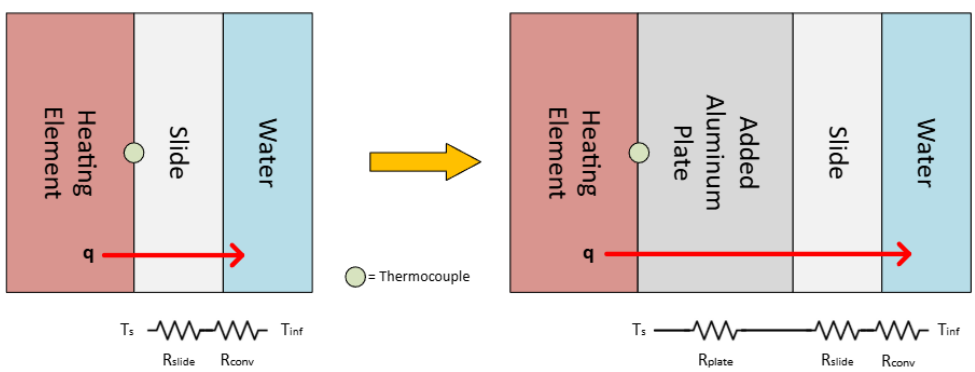

Figure 3.2: Thermal Resistance Circuits

$$R_{plate} = \Delta R = R_{total,plate} - R_{total,no\ plate}$$

Equation 3.3: Plate Thermal Resistance

The final parameter, cooling time of the heated surface, was found using careful examination of the test data. Consistency was maintained between test cases through the use of a predefined steady-state condition that was used for each of them.

This condition stated that the surface temperature was considered to be steady state when the change in temperature was less than 0.5 degrees Celsius over 20 seconds. This condition is validated by its use in earlier labs. The data was analyzed to determine where this occurred, and what the steady state temperature was. No calculations were necessary for this parameter, however, the test data is present in **Appendix C**. This data for each case has been cropped right after it reaches steady state.

4. Results and Discussion

Three parameters were examined in this experiment to effectively assess, understand, and analyze the cooling of a surface as a concept that may later be applied to practical systems such as battery cooling. The parameters tested were the effect on cooling time as different liquid coolants with varying salt concentrations were used, the influence of adding a plate and the thermal resistances introduced by additional surfaces on heat transfer and cooling rate, and the variation of the convection coefficient *h* with changes in salt concentration. The calculated values for each of these parameters are located in **Table 4.1** and **Table 4.2**. **Table 4.1** depicts the values for the initial surface temperature test case set at 50°C, while **Table 4.2** illustrates the test case for initial surface temperature at 80°C. The results and discussion below clearly explain the findings of the experiment.

Table 4.1 Final Parameter Data for 50°C initial surface temperature

Test case (°C-lb/gal-plate)	Cooling time (s)	*Ts* (°C)	T_∞(°C)	h (W/m²K)	R (K/W)
50-6lb-no	37	37.18	22.5	38.93	-
50-24lb-no	115	39.44	22.5	33.74	-
50-12lb-no	202	39.42	22.5	33.78	-
50-0lb-one	126	47.77	22.5	22.62	4.28
50-0lb-no	109	34.44	22.5	47.88	2.02

Table 4.2 Final Parameter Data for 80°C initial surface temperature

Test case (°C-lb/gal-plate)	Cooling time (s)	*Ts* (°C)	T_∞(°C)	h (W/m²K)	R (K/W)
80-6lb-no	247	49.12	22.5	65.50	-
80-24lb-no	154	56.74	22.5	50.93	-
80-12lb-no	223	57.08	22.5	50.43	-
80-0lb-one	170	75.65	22.5	32.81	2.95
80-0lb-no	140	42.96	22.5	85.24	1.14

Parameter 1: Cooling time

Figure 4.1 shows the temperature–time plot for the initial surface temperature of 50 °C. This plot was created to observe how cooling time changes as different cooling liquids are used. Ideally, it would be expected that as the salt concentration of the coolant increases, the cooling time would also increase. However, this trend did not appear. As shown in **Figure 4.1**, the cooling time for freshwater (green), 109 s, was much longer than the cooling time for the 0.5 lb/gal solution (blue), 37 s. This occurred because the experiment did not aim to cool the plate to a specific target temperature. Instead, the objective was to observe the lowest temperature each liquid reached upon achieving thermal equilibrium with its surroundings. Consequently, each coolant cooled the surface to a different final temperature, as illustrated in **Figure 4.1**. If the goal had been to cool the surface to a specific temperature, such as 15 °C, freshwater would indeed have reached that temperature faster than the other liquids.

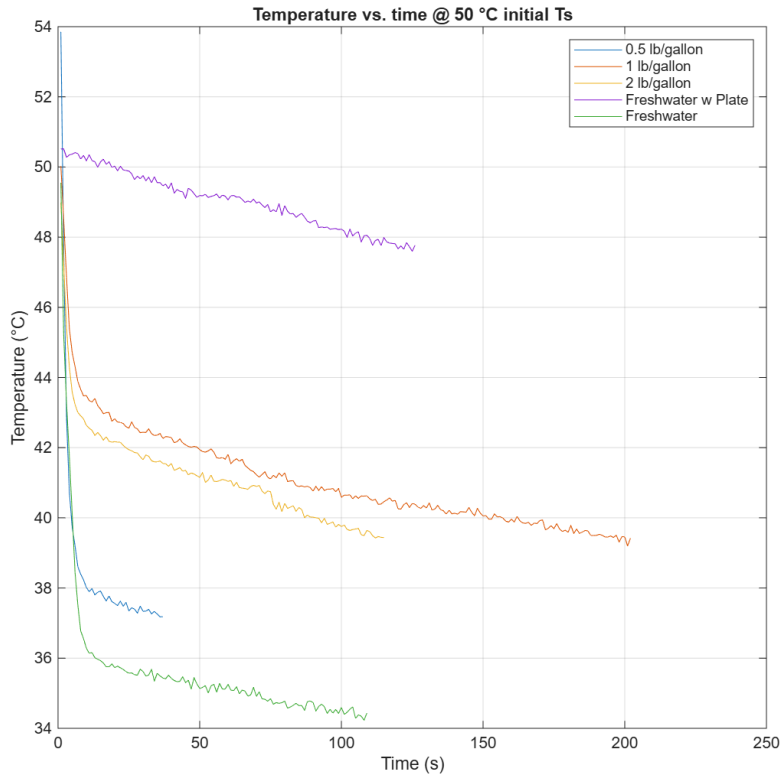

Figure 4.1 Temperature versus time plot for initial surface temperature set at 50 °C

The purple line at the top stands apart from the others and represents the plot for freshwater with an added plate. It is evident that the coolant with the plate inserted between the heater and the slide performs poorly, as it cools the surface only from 50 °C to 48 °C, as shown in **Figure 4.1**. This reduced effectiveness is due to the contact resistance caused by imperfect surface contact and the conduction resistance associated with the plate's thickness, both of which contribute to added thermal resistance that will be discussed later. **Table 4.1** presents the different liquid coolants, the final surface temperature each reached upon attaining thermal equilibrium with the surroundings, and the time required for each coolant to reach that final temperature.

It is important to note that no specific or consistent trend is evident regarding cooling time, other than the fact that all coolants were able to cool the slide surface. Additionally, a general conclusion such as cooling time increasing with salt concentration cannot be drawn here because each coolant was allowed to cool to a different final temperature.

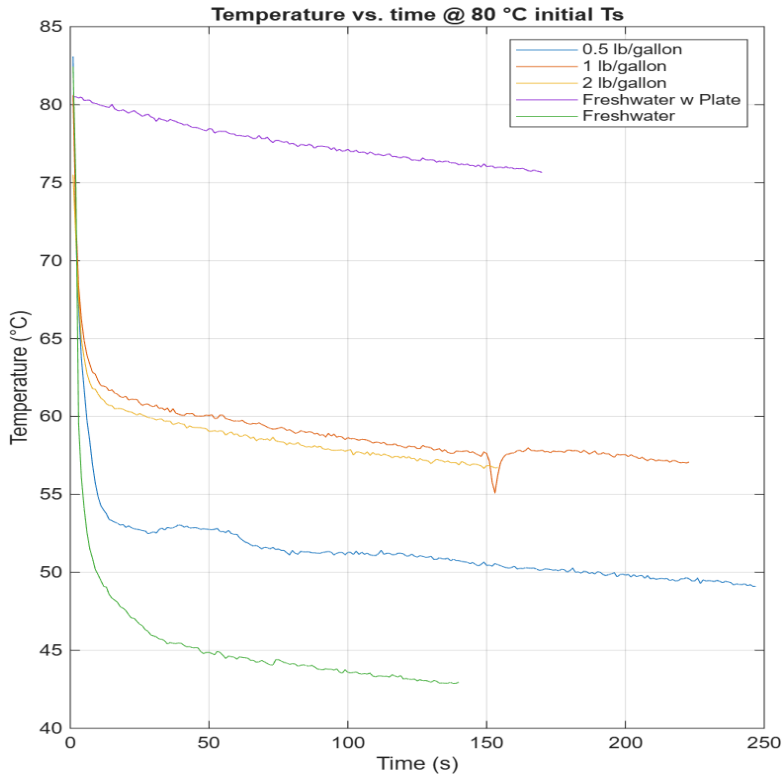

Figure 4.2 Temperature versus time plot for initial surface temperature set at 80 °C

Furthermore, **Figure 4.2** illustrates the temperature–time relationship for the different liquid coolants. The graph follows the expected general trend in which freshwater cools the surface to the lowest final temperature. As discussed earlier, because each coolant is allowed to cool until reaching thermal equilibrium with its surroundings, the coolants cool the surface to different final temperatures and require different amounts of time to do so. From the results, it is evident that the 0.5 lb/gal solution (blue) with a 5.7% salt concentration has the longest cooling time at 247 s, reaching a final steady-state temperature of 49 °C from an initial 80 °C.

In contrast, the coolant with the shortest cooling time is freshwater (green), which requires only 140 s to cool the surface from 80 °C to 43 °C. However, it is important to note that the cooling times of the various coolants cannot be directly compared because their final temperatures differ, making such a comparison impractical. For the 80 °C initial surface temperature, **Table 4.2** presents the liquid coolants, the final temperatures each reached upon attaining thermal equilibrium with the surroundings, and the corresponding cooling times.

An important phenomenon is observed in the temperature–time plot, where most of the cooling occurs at the very beginning of the experiment, typically within the first 20 seconds. Because water has a large specific heat capacity, it can absorb substantial amounts of heat with only a small change in temperature. At the start of the experiment, the large temperature gradient causes cooling to occur rapidly. As the surface temperature decreases, the temperature gradient between the surface and the liquid diminishes, reducing the rate of heat transfer. As the surface and liquid temperatures approach each other, cooling slows significantly, resulting in the plateau seen in the plot.

Parameter 2: Convection coefficient, *h*

The convection coefficient *h* is essentially a measure of a coolant's effectiveness. It is therefore fundamental to analyze, as it is the single parameter that most clearly indicates how well each coolant cools the slide surface. **Figures 3** and **4** show the relationship between *h* and salinity concentration, illustrating how increases in salt content affect cooling effectiveness. As seen in **Figure 3**, corresponding to the initial surface temperature of 50 °C, *h* decreases as salt concentration increases in the coolant. This trend aligns with expectations, since pure water is the most effective coolant.

However, a slight discrepancy appears in the results. As shown in **Figure 4.3**, when the salt concentration increased from 1 lb/gal (10.7% salt conc.) to 2 lb/gal (19.4% salt conc.), *h* did not change and remained at 33.78 W/m²K. Several factors may explain this inconsistency. The water may have been saturated, making additional salt ineffective. The mixture may not have been properly mixed, leading to a non-homogeneous solution, with salt particles visible either suspended or settled at the bottom and on the slide. Furthermore, since the mixing was performed manually rather than with precise equipment, inconsistencies were likely.

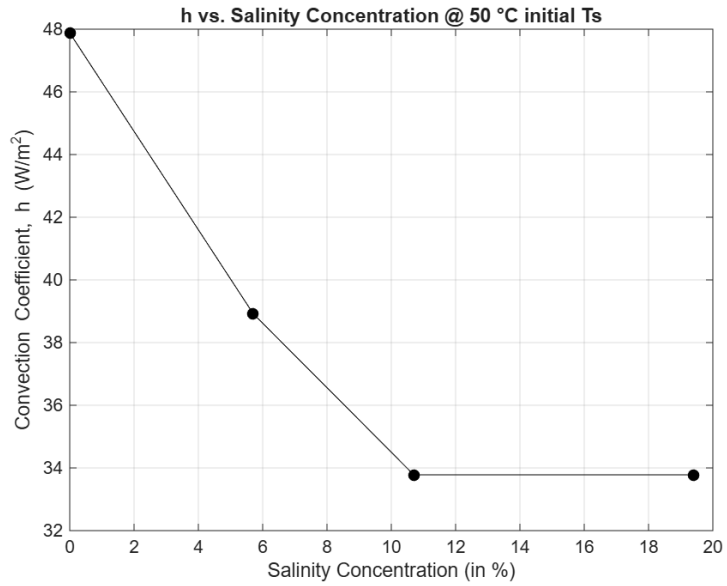

Figure 4.3 Convection coefficient vs. Salinity concentration for 50 °C test case

Similarly, for the initial surface temperature of 80 °C, a discrepancy was also observed as shown in **Figure 4.4** below. As the salt concentration increased from 1 lb/gal (10.7% salt conc.) to 2 lb/gal (19.4% salt conc.), *h* increased slightly from 50.43 to 50.93 W/m²K. In this case, the minimal increase may be explained by the possibility that accumulated salt particles on the slide surface enhanced heat transfer through conduction or convection and radiated heat to the environment, contributing to the slight rise in *h*.

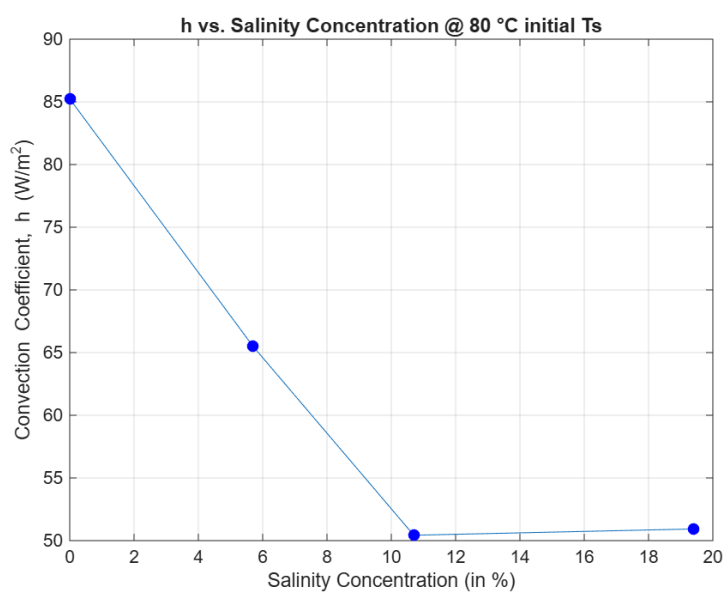

Figure 4.4 Convection coefficient vs. Salinity concentration for 80 °C test case

Despite the inconsistencies in the final two data points of both tests, the changes are minimal and can be considered insignificant. For example, an increase from 50.43 to 50.93 W/m²K represents only about a 1% change, which falls within reasonable experimental error.

Finally, it can be observed that, in general, the convection coefficients *h* for the 80 °C initial surface temperature are much higher than those for 50 °C. The values for the 50 °C condition ranged from 33.78 to 47.88 W/m²K, while those for 80 °C ranged from 50.43 to 85.24 W/m²K. This is because a larger amount of heat is available at higher temperatures, allowing more heat to be transferred and resulting in more effective cooling of the surface.

Parameter 3: Thermal Resistance, R_{th}

The results are summarized in the tables below, with **Table 4.3** corresponding to the 50 °C initial surface temperature and **Table 4.4** to the 80 °C initial surface temperature. Based on data analysis and calculations, it was found that the added thermal resistance of the plate was 1.81 K/W for the 80 °C test case and 2.26 K/W for the 50 °C test case.

There was always some thermal resistance between the heater and the slide because the thermocouple was placed between them, introducing imperfections in the contact between the surfaces. In addition, some thermal resistance exists from conduction through the slide and more exists from convection between the slide and the water flowing over it. This explains why thermal resistance is relatively significant even without the added plate. To compare the total thermal resistance introduced when inserting an aluminum plate between the heater and the slide, the liquid coolant was kept consistent, using freshwater in all cases. The initial surface temperatures were set to 50 °C and 80 °C, and the slide surface was cooled both with and without the added plate for each temperature condition.

Table 4.3 Thermal resistance for 50 °C

Test	R (K/W)	h (W/m²K)	Final *Ts* (°C)	Cooling time (s)
50C, no plate	2.02	47.88	34.44	109
50C, plate	4.28	22.62	47.77	126
Calculated Plate Resistance	2.26	-	-	-

To analyze the effects of adding an aluminum plate and how it influences cooling, **Table 4.3** compares the convection coefficient for the case with the added plate to the case without it with the initial *Ts* at 50 °C. The results show that the added plate significantly reduces the convection coefficient *h* from 47.88 to 22.62 W/m²K, representing nearly a 53% decrease in cooling effectiveness. This indicates that adding material, such as the plate in this case, generally introduces additional thermal resistance that impedes heat flow from the heater to the convection surface and reduces the fluid's cooling effectiveness by more than half. In the context of batteries, this would mean that adding extra material around the battery for insulation and safety would also mean a decrease in the capability of the battery's cooling system.

In addition, when comparing the cooling times and final temperatures after the liquid coolant reached thermal equilibrium with its environment, the 50 °C no-plate case cooled to about 34 °C, while the case with the plate cooled only to 48 °C. These results show that in practical applications, when effective heat transfer is required, minimizing the layers between conducting surfaces is essential, as additional materials introduce thermal resistances such as contact and conduction resistance that reduce overall heat flow.

The cooling time to reach and stabilize at 48 °C (126 s) was also longer than the time required to cool from 50 °C to 34 °C without the plate (109 s), further illustrating how an added plate or layer diminishes efficient heat transfer, as shown in **Table 4.3** above.

Table 4.4 Thermal resistance for 80 °C

Test	R (K/W)	h (W/m²K)	Final *Ts* (°C)	Cooling time (s)
80C, no plate	1.14	85.24	42.96	140
80C, plate	2.95	32.81	75.65	170
Calculated Plate Resistance	1.81	-	-	-

Equally important, it was necessary to observe how the associated thermal resistances changed under a higher temperature gradient. As summarized in **Table 4.4**, the system performed better at higher temperatures. The thermal resistance at 50 °C was 2.26 K/W, while at 80 °C it decreased to 1.81 K/W. As noted earlier, a larger temperature gradient promotes greater heat flow, which results in reduced thermal resistance.

For the 80 °C initial surface temperature, the convection coefficient h was 85.24 W/m²K and dropped to 32.81 W/m²K when the plate was added, representing a 61% reduction in cooling effectiveness. With the plate, the final surface temperature cooled only to about 76 °C, compared to 43 °C without the plate. Additionally, it took longer (170 s) for the coolant to cool the surface with the plate to 76 °C than it took to cool the surface to 43 °C without the plate (140 s), further reinforcing the negative impact of adding plates or barriers between surfaces where effective heat transfer is required.

Experimental Error

There were several sources of error in the experiment. The first of which was the variation of flow rate between test cases. While the flow rate was set at a constant 0.84 gallons per minute (gpm), during some cases this varied by as much as 0.04 gpm. Convective heat transfer, and by extension convection coefficient, is proportional to the flow rate of the convective fluid.

Therefore **Equation 4.1** can be used to determine that the flow rate variation could result in an error of as much as 5.0% in the calculated convection coefficient.

This variation was more prominent in test cases with a greater salt content. The increased viscosity of the fluid from the additional salt is likely the cause of the discrepancy, as it could have slowed the pump.

$$Error = \left|\frac{Measured - Actual}{Actual}\right| \times 100\%$$

Equation 4.1: Percent Error Equation

Another potential source of error was improper mixing of the cooling liquid. The liquid was stirred with a pipe in a large box, as there was not a better way to properly mix the solution in this laboratory setting. It was previously discussed that this likely had an impact on the obtained convection coefficients for the test cases that utilized 2 lb/gal (19.4% concentration) of salt.

Using the assumption that this was the only cause of this test case varying dramatically from the otherwise linear trend of the others, **Figure 4.3** and **Figure 4.4** can be used to determine that this anomaly could have caused a percent error of up to 71.6% for the 50°C 2 lb/gal test case, and up to 206.3% for the 80°C 2 lb/gal test case. It is important to keep in mind that while these percentage error values are very high, they are only applicable to one of the four solutions.

The final quantifiable source of error was due to the thermal resistance of the slide. While the slide does have thermal resistance, it was assumed that this value was relatively small. Because of this, surface temperature was measured from just under the slide in all test cases except for the ones with the added plate. The added plate and the slide are both comprised of aluminum, with the plate being 0.050 inches thick and the slide being 0.025 inches thick. If we neglect the effect of contact resistance, we can get a rough estimate for the added thermal resistance of the slide based on the calculated thermal resistance of the added plate. Since conduction resistance is proportional to the thickness of a material, the resistance of the slide can be estimated to be half of the resistance of the plate, or 0.91 K/W for the 80 °C test cases and 1.13 K/W for the 50 °C test cases.

From the overall system resistivity of 1.14 K/W and 2.02 K/W for 80 degrees Celsius and 50 degrees Celsius respectively, it can be determined that this resistance could have resulted in a maximum error of 395.7% and 127.0% in thermal resistance and overall heat transfer for 80°C and 50°C respectively. While these numbers are also very high, they neglect the contact resistance of adding the aluminum plate and assume that the entire added resistance is from conduction. These errors will be much lower as the resistivity of the slide does not include adding the additional contact resistances that the plate adds to the system. In addition, this error is uniform throughout all test cases and therefore does not affect the comparison of results between them.

Improvements

One major issue with this experiment was insulation. There was nothing to stop heat from travelling through the slide and escaping away from the insulation. In the calculations, the setup was approximated to have 1D heat transfer.

Since the aluminum slide was thin, the approximation wasn't as inaccurate, but the setup could have been improved to increase the accuracy of the approximation. More insulation could have been added to completely cover the sides and bottom of the slide.

Also, the 4x4 inch square area on the bottom of the slide that the heating pad was connected to could have been cut out and glued back in place, such that the glue insulated the plate from the rest of the slide.

This experiment also had several issues with regards to data collection, including with the measured convection surface, the measured fluid flow rate, and the assumed homogenous coolant solution. To prevent thermal resistance from interfering with the measurement for surface temperature, as stated above, thermocouples should be embedded in the material of the slide itself. This could be done by using a thicker material and cutting slots to insert the devices. This would potentially increase thermal resistance from conduction, but would virtually eliminate any contact resistances. Issues with the fluid flow rate could be solved by using a pump with greater flow adjustability. The pump used in lab simply had a high speed and low speed feature, which is helpful for general flow adjustments but lacks the precision to set an exact flow rate. Finally, the issue caused by improper mixing of the solution could be solved by using special equipment to help with the task, such as a laboratory grade stirring machine.

This experiment also points at several improvements that can be made in battery technology as a whole. The thermal resistance calculations showed that adding extra material around a battery would significantly impact the effectiveness of liquid cooling on it. Therefore it is important to optimize the design of battery shielding and insulation to use as little material as possible, ensuring the most effective heat transfer with the cooling fluid. In addition, the experiment showed that diluting effective cooling liquids with various additives can significantly decrease the cooling capability of the fluids. There are multiple different types of EV cooling liquids, and several different additives that can be mixed into these for various reasons (Tone Cooling, 2025). To prevent decreased cooling capability, it is important to utilize the purest cooling liquids possible in battery cooling systems with as few additives as possible, which the experiment indicates will best optimize steady state temperature and cooling time.

5. Conclusions

This experiment investigated the key engineering challenge of thermal management for electric vehicle batteries by testing the performance of different liquid coolants.

A lab-scale setup was built, including a heated metal plate simulating a battery module and a controlled flow of temperature-regulated liquid for cooling. The objective was to measure and evaluate the cooling efficiency of pure water and brine solutions using fundamental heat transfer analysis.

Through the successful implementation of experimental and analytical procedures, this experiment achieved many major accomplishments.

- Successfully designing, iterating, and building a functional experimental setup to investigate liquid convection cooling.
- Systematically collecting transient temperature data for various coolants and operating conditions using a full instrumentation suite.
- Calculating and comparing key performance metrics, the convective heat transfer coefficient (h), thermal resistance (R_{th}), and cooling dynamics for each test case.
- Quantifying the considerable performance penalty caused by increased heat contact and conduction resistance.
- Conclusively determining that pure water provides superior cooling performance compared to the tested saltwater brines for this application.

The results indicate that the project's objectives were met. It was determined that pure water is the most effective coolant, with the maximum convection coefficient (h). The analysis also showed that thermal resistance (R_{th}) has a considerable impact on the effective convection coefficient (h); the extra plate reduced h by more than 50%, emphasizing the importance of interface design. Furthermore, the analysis highlighted that cooling time is integrally linked to both the initial and final goal temperatures, making the convection coefficient a more reliable value for comparison.

Based on the experimental and analytical results, worthwhile extensions of this work would include testing industry-standard coolants such as ethylene and propylene glycol (Dober, 2025) while adhering to strict safety protocols for handling chemical fluids, investigating the influence of various flow geometries (e.g., internal channels). Future work could also systematically explore how variations in coolant flow rate affect the convection coefficient in order to optimize the trade-off between cooling performance and the required pumping power. Implementing more complete insulation to better satisfy the 1D heat transfer assumption might increase future iteration accuracy. In conclusion, for direct liquid cooling applications where fluid characteristics are critical, our study verifies that pure water is a superior coolant to the alternatives tested and emphasizes the importance of minimizing parasitic thermal resistances in system design.

6. References

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3. Tone Cooling Technology Co., Ltd. (2025, November 29). *EV battery cooling methods: Air, liquid and direct refrigerant explained*. tonecooling.com. https://tonecooling.com/ev-battery-cooling-methods/
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7. Appendix

Appendix A: Convective Heat Transfer Coefficient Derivation

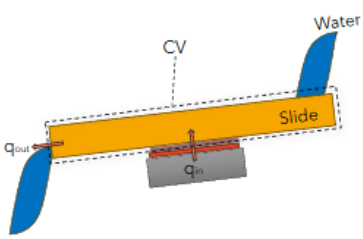
Assumptions:

- 1-D Heat Transfer
- Insulated Sides
- Negligible Radiation
- Negligible Temperature Rise Within Fluid

Terms:

- $\dot{E}$: Energy Transfer Rate (W)
- q: Heat Transfer Rate (W)
- h: Convective Heat Transfer Coefficient (W/m^2K)
- T: Temperature (K)
- A: Surface Area (m^2)

Schematic:



Analysis:

Starting with an energy balance and solving for the convection coefficient, h

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_{gen} = \dot{E}_{st}$$

$$\dot{E}_{st} = 0 \qquad \dot{E}_{gen} = 0$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$q_{in} = q_{out}$$

$$q_{in} = h(T_s - T_\infty)$$

$$h = \frac{q}{A(T_s - T\infty)}$$

Sample Calculation:

80 degrees Celsius starting temperature, pure water, no added plate test case:

$$q = 18.0\ watts$$

$$A = 0.010\ meters^2$$

$$T_\infty = 22.5\ degree\ Celsius$$

$$T_s = 42.96\ degree\ Celsius$$

$$h = \frac{q}{A(T_s - T\infty)}$$

$$h = \frac{18.0}{0.010(42.96 - 22.5)}$$

$$h \approx 85.24\ \frac{W}{m^2 K}$$

Appendix B: Thermal Resistance Derivation

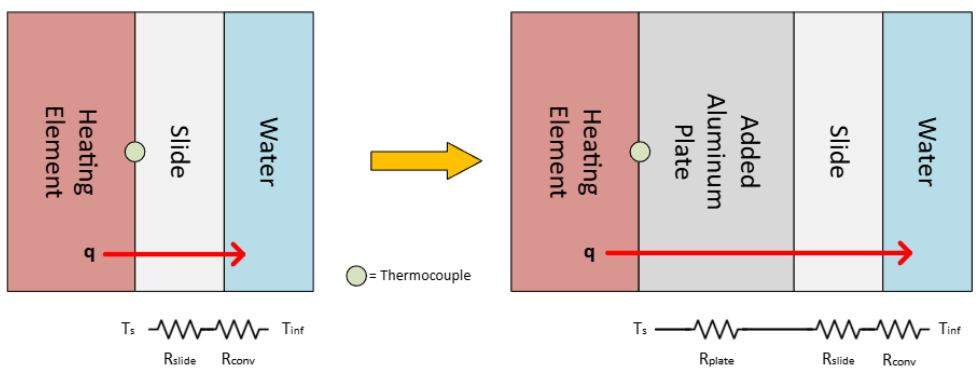
Assumptions:

- 1-D Heat Transfer
- Insulated Sides
- Negligible Radiation
- Negligible Temperature Rise Within Fluid

Terms:

- q: Heat Transfer Rate (W)
- R: Thermal Resistance (K/W)
- T: Temperature (K)

Schematic:



Analysis:

Start by solving for thermal resistance, R_{th}, then relating the plate's thermal resistance to the difference between the total system thermal resistance with and without the plate

$$q = \frac{T_s - T_\infty}{R}$$

$$R_{total} = \frac{T_s - T_\infty}{q}$$

$$R_{plate} = \Delta R = R_{total,plate} - R_{total,no\ plate}$$

Sample Calculation:

80 degrees Celsius starting temperature, pure water test case:

$$q = 18.0\ watts$$

$$T_\infty = 22.5\ degree\ Celsius$$

With Plate

$$T_s = 75.65\ degree\ Celsius$$

Without Plate

$$T_s = 42.96\ degree\ Celsius$$

$$R_{total,plate} \approx \frac{75.65 - 22.5}{18}$$

$$R_{total,plate} = 2.95\ \frac{K}{w}$$

$$R_{total,no\ plate} = \frac{42.96 - 22.5}{18}$$

$$R_{total,no\ plate} \approx 1.14\ \frac{K}{w}$$

$$R_{plate} = \Delta R = R_{total,plate} - R_{total,no\ plate}$$

$$R_{plate} = 2.95\ -\ 1.14$$

$$R_{plate} = 1.81\ \frac{K}{W}$$

Appendix C: Test Data

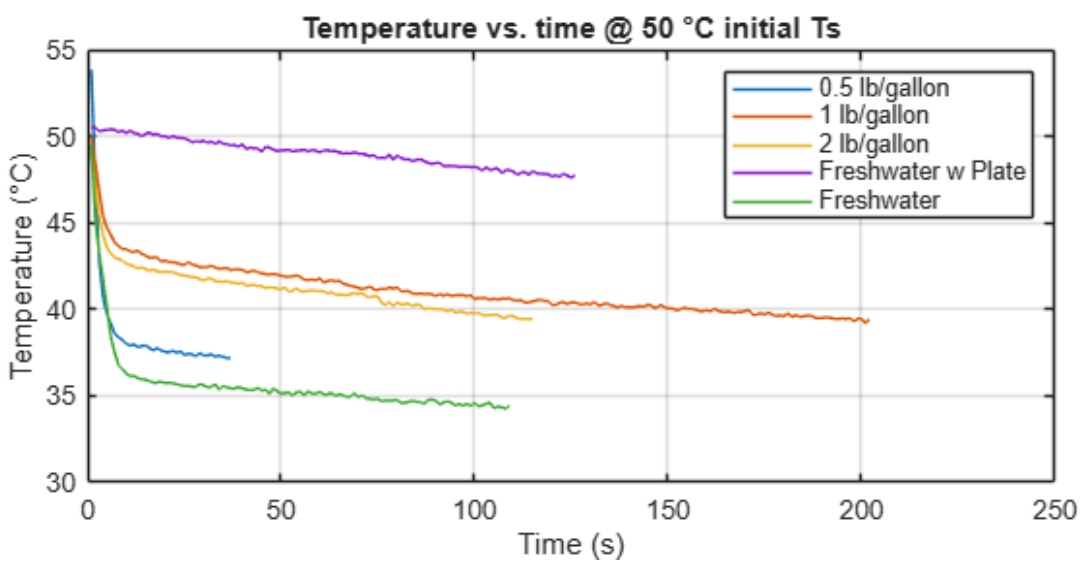
The data collected during the experiment that was used for data analysis and calculations is attached at the end of this report as well as the MATLAB code utilized to generate figures.

```
clc;
close all;
clear;

data = readmatrix("Combined_data.xlsx", "Sheet", "T_vs_t_50");

time = data(:,1);
lb_6 = data(:,2);
lb_24 = data(:,3);
lb_12 = data(:,4);
freshwater_plate = data(:,5);
freshwater = data(:,6);

% Plot the data for visualization
figure;
plot(time, lb_6, time, lb_12, time, lb_24, time, freshwater_plate, time,
freshwater);
xlabel('Time (s)');
ylabel('Temperature (°C)');
legend('0.5 lb/gallon', '1 lb/gallon', '2 lb/gallon', 'Freshwater w Plate',
'Freshwater');
title('Temperature vs. time @ 50 °C initial Ts');
grid on;
```



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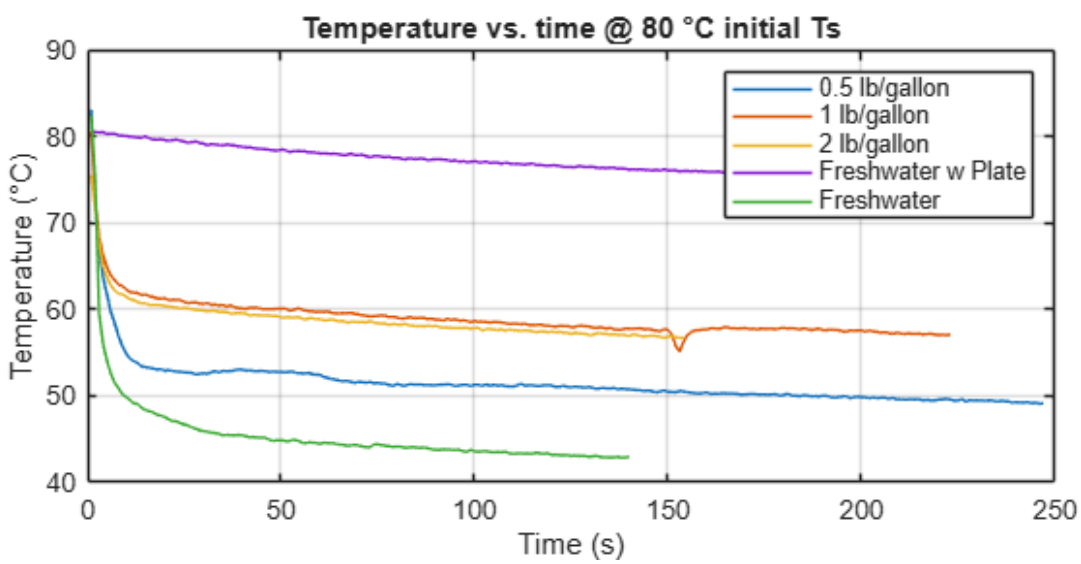
```
clc;
close all;
clear;

data = readmatrix("Combined_data.xlsx", "Sheet", "T_vs_t_80");

time = data(:,1);
lb_6 = data(:,2);
lb_24 = data(:,3);
lb_12 = data(:,4);
freshwater_plate = data(:,5);
freshwater = data(:,6);

% Plot the data for visualization
figure;
plot(time, lb_6, time, lb_12, time, lb_24, time, freshwater_plate, time,
freshwater);
xlabel('Time (s)');
ylabel('Temperature (°C)');
legend('0.5 lb/gallon', '1 lb/gallon', '2 lb/gallon', 'Freshwater w Plate',
'Freshwater');
title('Temperature vs. time @ 80 °C initial Ts');
grid on;
```



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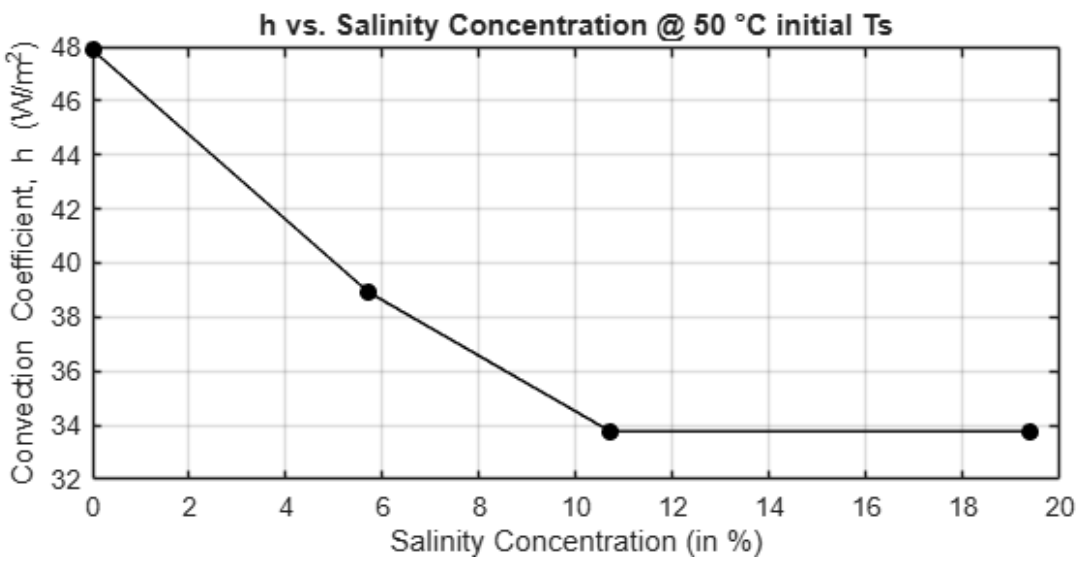
```
clear;
close all;
clc;

data = readmatrix("cooling_data", "Sheet", "T50");

salinity = data(:,7);
h = data(:,6);

figure;
plot(salinity, h,'Color','k');
hold on;
plot(salinity,
h,'o','MarkerSize',6,'MarkerFaceColor','k','MarkerEdgeColor','k');
xlabel('Salinity Concentration (in %)');
ylabel('Convection Coefficient, h (W/m^2)');
%ylabel('Convection Coefficient, $h\ (W/m^{2})$', 'Interpreter', 'latex')
title('h vs. Salinity Concentration @ 50 °C initial Ts');
grid on;
```



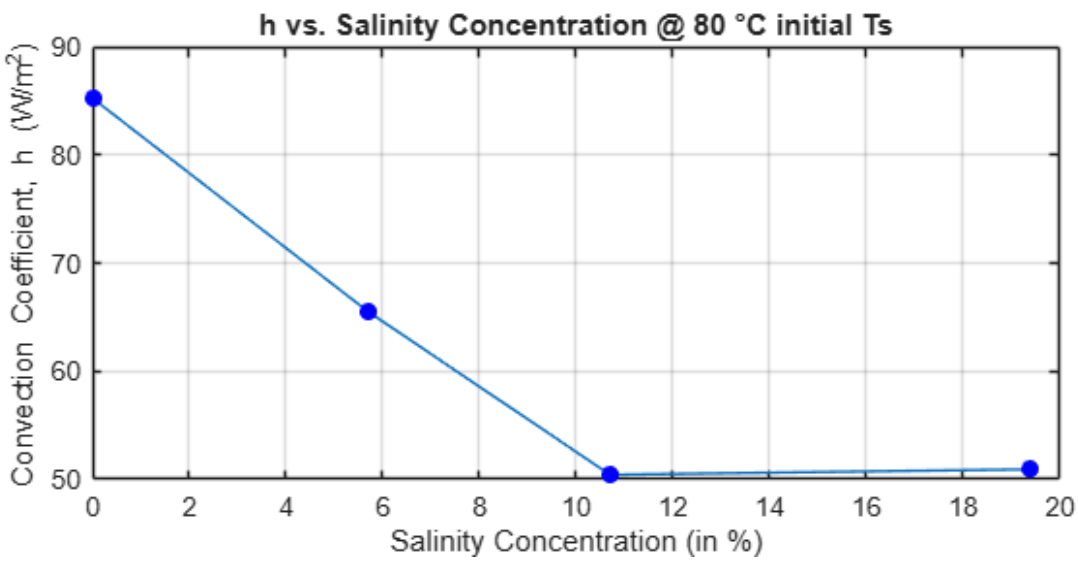
Published with MATLAB® R2025b

```
clear;
close all;
clc;

data = readmatrix("cooling_data", "Sheet", "T80");

salinity = data(:,7);
h = data(:,6);

figure;
plot(salinity, h);
hold on;
plot(salinity,
h,'o','MarkerSize',6,'MarkerFaceColor','b','MarkerEdgeColor','b');
xlabel('Salinity Concentration (in %)');
ylabel('Convection Coefficient, h (W/m^2)');
%ylabel('Convection Coefficient, $h\ (W/m^{2})$', 'Interpreter', 'latex')
title('h vs. Salinity Concentration @ 80 °C initial Ts');
grid on;
```



Published with MATLAB® R2025b

test	Cooling tim	Ts	Tinf	h	r
80-6lb-no	247	49.12252	22.5	65.5	
80-24lb-no	154	56.73582	22.5	50.93	
80-12lb-no	223	57.07863	22.5	50.43	
80-0lb-one	170	75.6517	22.5	32.81	2.95
80-0lb-no	140	42.95658	22.5	85.24	1.14
50-6lb-no	37	37.18363	22.5	38.93	
50-24lb-no	115	39.43806	22.5	33.78	
50-12lb-no	202	39.42123	22.5	33.78	
50-0lb-one	126	47.77015	22.5	22.62	4.28
50-0lb-no	109	34.43652	22.5	47.88	2.02

time	50-6lb-no	50-24lb-no	50-12lb-no	50-0lb-one	50-0lb-no
1	53.85236	48.98482	50.02561	50.51362	49.55157
2	47.59708	47.21564	48.49025	50.51514	45.20186
3	42.84901	45.41606	46.84975	50.27681	43.27471
4	40.72733	44.2922	45.39487	50.3463	41.75646
5	39.73398	43.58456	44.70748	50.36607	40.11266
6	39.20355	43.23009	44.31273	50.40898	38.44718
7	38.60373	43.00321	43.88431	50.38343	37.50948
8	38.39388	42.90755	43.66258	50.22715	36.7788
9	38.23135	42.81878	43.47626	50.33077	36.5536
10	38.01462	42.63549	43.48802	50.17677	36.28608
11	37.8973	42.55418	43.34124	50.34855	36.13822
12	37.98312	42.4949	43.30208	50.18116	36.15903
13	37.80211	42.34771	43.40104	50.15122	36.01338
14	37.87975	42.43266	43.1973	49.99495	35.96805
15	37.91204	42.34158	43.09767	50.14578	35.9299
16	37.75856	42.21404	42.96979	50.22194	35.8604
17	37.63104	42.31186	42.99609	50.07522	35.75524
18	37.77208	42.18214	43.0039	50.15024	35.75662
19	37.6134	42.15649	42.75573	49.98714	35.83845
20	37.56099	42.17217	42.8211	50.02687	35.73153
21	37.49976	42.15795	42.72965	49.89313	35.77232
22	37.62292	42.15629	42.7234	50.01974	35.72582
23	37.48258	42.06522	42.68855	49.91785	35.67983
24	37.58742	41.99365	42.59658	49.89985	35.61508
25	37.34781	41.94048	42.55583	49.88351	35.57194
26	37.43392	41.91462	42.73434	49.80488	35.57951
27	37.39918	41.86075	42.57883	49.63963	35.52989
28	37.28435	41.85002	42.53425	49.73842	35.51462
29	37.48287	41.74291	42.4269	49.66431	35.68442
30	37.33549	41.65667	42.43842	49.7554	35.62019
31	37.34056	41.79054	42.43352	49.60683	35.49176
32	37.39104	41.75565	42.53712	49.70946	35.51748
33	37.26275	41.62691	42.39813	49.54956	35.68416
34	37.33241	41.59541	42.34974	49.72115	35.34325
35	37.25739	41.59318	42.35514	49.5548	35.5697
36	37.17657	41.62192	42.40494	49.55434	35.49418
37	37.18363	41.54892	42.26439	49.46539	35.43346
38		41.5405	42.31519	49.50893	35.41395
39		41.47044	42.30786	49.38176	35.51608
40		41.54692	42.28594	49.54679	35.42532
41		41.35277	42.14238	49.25467	35.36363
42		41.38704	42.16982	49.35811	35.32863
43		41.43097	42.25044	49.30949	35.34042

44	41.33018	42.1335	49.29116	35.46933
45	41.35293	42.08261	49.10569	35.30245
46	41.22429	42.0327	49.39562	35.37334
47	41.27988	42.01576	49.33859	35.12276
48	41.24795	42.03541	49.24276	35.44392
49	41.20135	42.01961	49.13546	35.26641
50	41.15153	41.94312	49.18771	35.13497
51	41.28953	41.8909	49.17885	35.1674
52	41.00935	41.87113	49.22194	35.21636
53	41.07594	41.90758	49.13107	35.01193
54	41.2188	41.96401	49.16294	35.20714
55	41.043	41.87162	49.1844	35.25649
56	41.0344	41.7104	49.22807	35.02791
57	41.05908	41.69801	49.11188	35.28037
58	41.09511	41.71896	49.20158	35.11715
59	41.07517	41.67	49.16753	35.12033
60	41.03872	41.80865	49.06465	35.25291
61	41.0625	41.54273	49.18033	35.08925
62	40.90076	41.62954	49.18785	35.07736
63	41.01152	41.68489	49.14266	35.18413
64	40.86297	41.61315	49.14611	34.98419
65	40.84714	41.66673	49.07674	35.09228
66	40.8347	41.49349	48.991	35.05875
67	40.81374	41.42345	49.0288	34.91827
68	40.89166	41.35226	48.99897	34.95404
69	40.92657	41.31642	49.08192	35.16887
70	40.89507	41.22777	48.97415	34.91156
71	40.92113	41.15938	49.0063	35.03889
72	40.76417	41.23534	48.91746	34.84191
73	40.69044	41.31328	48.84486	34.82878
74	40.76519	41.15212	48.9281	34.77361
75	40.74333	41.11261	48.72223	34.8432
76	40.38167	41.21263	48.78343	34.6887
77	40.2475	41.14438	48.72319	34.73353
78	40.44059	41.27779	48.95266	34.71109
79	40.17242	41.18295	48.62504	34.7319
80	40.41398	41.28463	48.9005	34.77488
81	40.26962	41.0357	48.74078	34.57661
82	40.30487	41.05308	48.6615	34.60669
83	40.34484	41.05646	48.67364	34.6553
84	40.18729	40.92039	48.56639	34.71202
85	40.28975	40.8917	48.62928	34.64737
86	40.23957	40.91086	48.67621	34.6506
87	39.999	40.88202	48.57276	34.51325

88	40.07952	40.89961	48.4559	34.76102
89	40.02212	40.86145	48.41158	34.77991
90	40.01885	40.76001	48.45998	34.73537
91	39.99986	40.89705	48.47145	34.49282
92	39.98854	40.78019	48.26944	34.6235
93	39.86203	40.88776	48.28959	34.68886
94	39.98494	40.77821	48.27263	34.61367
95	39.81563	40.81912	48.287	34.42491
96	39.84386	40.81795	48.22612	34.54185
97	39.88064	40.83567	48.23986	34.43427
98	39.73598	40.74616	48.24523	34.53965
99	39.81075	40.83684	48.21703	34.42383
100	39.75204	40.58357	48.22801	34.59472
101	39.79509	40.64812	48.17254	34.39959
102	39.67245	40.64595	47.99952	34.45533
103	39.63942	40.65649	48.22946	34.54133
104	39.6677	40.54374	48.03509	34.61289
105	39.68003	40.627	48.11222	34.28396
106	39.70058	40.5544	48.15427	34.3854
107	39.51776	40.6249	47.86249	34.34758
108	39.49531	40.61543	48.04081	34.23242
109	39.64021	40.63041	48.0527	34.43652
110	39.60644	40.54017	47.96348	
111	39.49331	40.49708	47.7666	
112	39.42682	40.52786	47.90499	
113	39.46541	40.38505	47.93792	
114	39.4448	40.39321	47.76121	
115	39.43806	40.45873	47.98627	
116		40.50759	47.86767	
117		40.56854	47.83052	
118		40.47559	47.82025	
119		40.49499	47.80913	
120		40.25667	47.6696	
121		40.25049	47.74777	
122		40.37411	47.65681	
123		40.42897	47.84643	
124		40.29562	47.72936	
125		40.40843	47.59901	
126		40.38984	47.77015	
127		40.30758		
128		40.27181		
129		40.34906		
130		40.28116		
131		40.42494		

132	40.21799
133	40.25388
134	40.35557
135	40.23458
136	40.19958
137	40.10696
138	40.21517
139	40.11473
140	40.13024
141	40.10312
142	40.17593
143	40.16965
144	40.162
145	40.29321
146	40.15692
147	40.05799
148	40.26886
149	40.13647
150	40.05505
151	40.06314
152	39.96376
153	39.9612
154	40.1102
155	40.02485
156	40.0337
157	39.99299
158	39.90681
159	39.87854
160	39.96111
161	39.88021
162	40.03852
163	39.9149
164	39.85381
165	39.84223
166	39.87042
167	39.7855
168	39.84816
169	39.83732
170	39.91968
171	39.93974
172	39.65689
173	39.70156
174	39.7758
175	39.69199

176	39.8356
177	39.68075
178	39.61103
179	39.65002
180	39.58903
181	39.78267
182	39.55353
183	39.68023
184	39.5573
185	39.59505
186	39.63921
187	39.62605
188	39.49631
189	39.49681
190	39.49972
191	39.52947
192	39.50394
193	39.45651
194	39.45141
195	39.48288
196	39.43947
197	39.50523
198	39.31
199	39.46431
200	39.45574
201	39.19507
202	39.42123
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time	80-6lb-no	80-24lb-no	80-12lb-no	80-0lb-one	80-0lb-no	50-6lb-no	50-24lb-no	50-12lb-no
1	83.08389	75.50306	80.59431	80.51164	82.43108	53.85236	48.98482	50.02561
2	71.47997	71.52645	72.70062	80.49705	75.00033	47.59708	47.21564	48.49025
3	67.12911	67.62328	68.34239	80.4343	59.52663	42.84901	45.41606	46.84975
4	63.74218	64.95389	66.23718	80.48662	56.08549	40.72733	44.2922	45.39487
5	61.74124	63.75688	64.86832	80.26502	54.10888	39.73398	43.58456	44.70748
6	59.77226	62.77292	63.94436	80.28624	52.58899	39.20355	43.23009	44.31273
7	58.45431	62.18761	63.36819	80.28463	51.52706	38.60373	43.00321	43.88431
8	56.93119	61.79164	62.84038	80.11238	50.90488	38.39388	42.90755	43.66258
9	55.7529	61.74977	62.71706	80.08166	50.20295	38.23135	42.81878	43.47626
10	54.86937	61.4536	62.29799	80.02654	49.85788	38.01462	42.63549	43.48802
11	54.29494	61.18432	61.98604	79.95214	49.49355	37.8973	42.55418	43.34124
12	54.00849	61.05427	61.94116	79.90837	49.11716	37.98312	42.4949	43.30208
13	53.76945	60.81604	61.88215	79.85223	49.05843	37.80211	42.34771	43.40104
14	53.39593	60.70502	61.64365	79.82375	48.59861	37.87975	42.43266	43.1973
15	53.31891	60.69448	61.71342	80.00997	48.35007	37.91204	42.34158	43.09767
16	53.23318	60.49282	61.50845	79.73368	48.22598	37.75856	42.21404	42.96979
17	53.14976	60.51064	61.42909	79.62788	48.08254	37.63104	42.31186	42.99609
18	53.04541	60.50828	61.26619	79.60269	47.80657	37.77208	42.18214	43.0039
19	53.07641	60.46606	61.22367	79.67374	47.70959	37.6134	42.15649	42.75573
20	52.89142	60.37617	61.29943	79.56885	47.56027	37.56099	42.17217	42.8211
21	52.98395	60.31277	61.05479	79.45523	47.29522	37.49976	42.15795	42.72965
22	52.87621	60.2773	61.10579	79.54822	47.07662	37.62292	42.15629	42.7234
23	52.80353	60.19268	61.08782	79.61256	47.05495	37.48258	42.06522	42.68855
24	52.76361	60.09154	61.0908	79.47234	46.86803	37.58742	41.99365	42.59658
25	52.73283	60.19095	60.86065	79.37517	46.67602	37.34781	41.94048	42.55583
26	52.67908	60.0914	60.7102	79.24767	46.57964	37.43392	41.91462	42.73434
27	52.66195	60.02362	60.79126	79.33009	46.32359	37.39918	41.86075	42.57883
28	52.4826	59.92625	60.80337	79.39095	46.21866	37.28435	41.85002	42.53425
29	52.53122	59.92264	60.67693	79.21676	45.99151	37.48287	41.74291	42.4269
30	52.61409	59.85277	60.61247	79.1569	45.92932	37.33549	41.65667	42.43842
31	52.52482	59.77507	60.73598	78.93475	45.86073	37.34056	41.79054	42.43352
32	52.74988	59.83181	60.51355	79.13025	45.73589	37.39104	41.75565	42.53712
33	52.84699	59.82926	60.54939	79.02785	45.70792	37.26275	41.62691	42.39813
34	52.72229	59.68787	60.4122	79.0118	45.61059	37.33241	41.59541	42.34974
35	52.8874	59.70755	60.54482	78.92103	45.41027	37.25739	41.59318	42.35514
36	52.84256	59.68927	60.2644	79.04887	45.50405	37.17657	41.62192	42.40494
37	52.86549	59.48569	60.49631	78.98102	45.45444	37.18363	41.54892	42.26439
38	52.94046	59.53953	60.21648	78.92635	45.43791		41.5405	42.31519
39	53.05257	59.59809	60.15518	78.82936	45.47378		41.47044	42.30786
40	53.02345	59.53154	60.0864	78.79022	45.4514		41.54692	42.28594
41	52.90095	59.49996	60.06095	78.69605	45.32234		41.35277	42.14238
42	52.96267	59.25883	60.20208	78.7285	45.22052		41.38704	42.16982
43	52.85445	59.39257	60.17523	78.62522	45.24444		41.43097	42.25044

44	52.86088	59.3124	60.19833	78.53963	45.15964	41.33018	42.1335
45	52.87995	59.28871	60.02904	78.54318	45.17262	41.35293	42.08261
46	52.84349	59.29148	60.02268	78.50948	45.16398	41.22429	42.0327
47	52.76409	59.27951	60.01761	78.43477	44.92711	41.27988	42.01576
48	52.80004	59.23994	60.02535	78.37156	44.79714	41.24795	42.03541
49	52.77782	59.2054	60.0827	78.32422	44.87438	41.20135	42.01961
50	52.75899	59.04733	60.05379	78.45843	44.87372	41.15153	41.94312
51	52.73293	59.0543	60.12237	78.43154	44.80229	41.28953	41.8909
52	52.76606	59.09106	59.86392	78.17557	44.71924	41.00935	41.87113
53	52.82094	59.02946	59.89423	78.26935	44.94912	41.07594	41.90758
54	52.61141	59.09097	60.09709	78.28994	44.86271	41.2188	41.96401
55	52.63088	59.04139	60.0758	78.27903	44.68263	41.043	41.87162
56	52.5542	58.80873	59.86393	78.28389	44.48673	41.0344	41.7104
57	52.69277	58.93186	59.81123	78.17272	44.62921	41.05908	41.69801
58	52.4919	59.00082	59.76444	78.1221	44.59464	41.09511	41.71896
59	52.44648	58.83474	59.70214	78.12633	44.56646	41.07517	41.67
60	52.44646	58.89244	59.72486	77.99774	44.44318	41.03872	41.80865
61	52.22045	58.72959	59.70065	78.0185	44.58467	41.0625	41.54273
62	52.07636	58.73645	59.75753	77.98936	44.54672	40.90076	41.62954
63	51.94968	58.83357	59.61	77.96258	44.5291	41.01152	41.68489
64	51.88208	58.74001	59.53804	77.99933	44.4977	40.86297	41.61315
65	51.77481	58.59527	59.51112	77.90662	44.35333	40.84714	41.66673
66	51.68518	58.681	59.56506	77.81396	44.35496	40.8347	41.49349
67	51.72757	58.45069	59.53288	77.83966	44.24015	40.81374	41.42345
68	51.62977	58.54961	59.59687	78.01935	44.2712	40.89166	41.35226
69	51.50396	58.56743	59.48702	77.81425	44.34752	40.92657	41.31642
70	51.62602	58.50529	59.40655	77.81869	44.26613	40.89507	41.22777
71	51.58417	58.562	59.31309	77.90255	44.15086	40.92113	41.15938
72	51.5154	58.461	59.23864	77.70443	44.07237	40.76417	41.23534
73	51.61204	58.66763	59.18805	77.56294	44.06026	40.69044	41.31328
74	51.46659	58.44499	59.19725	77.70844	44.38968	40.76519	41.15212
75	51.49491	58.39211	59.24373	77.71143	44.40257	40.74333	41.11261
76	51.41199	58.36136	59.30395	77.63049	44.3183	40.38167	41.21263
77	51.41286	58.35481	59.2423	77.58221	44.22885	40.2475	41.14438
78	51.25933	58.18647	59.10726	77.63275	44.1854	40.44059	41.27779
79	51.12735	58.31444	59.15553	77.47592	44.10609	40.17242	41.18295
80	51.39847	58.23791	59.04697	77.52902	44.13918	40.41398	41.28463
81	51.23025	58.2154	58.92801	77.44036	44.12524	40.26962	41.0357
82	51.1608	58.33468	58.95846	77.31242	43.98191	40.30487	41.05308
83	51.34821	58.16311	59.08637	77.43718	44.04102	40.34484	41.05646
84	51.31647	58.11816	59.0387	77.42342	44.07689	40.18729	40.92039
85	51.30866	58.04516	58.96974	77.35314	44.02329	40.28975	40.8917
86	51.32136	58.07225	59.01881	77.44981	43.98152	40.23957	40.91086
87	51.24823	58.02979	58.99418	77.38478	44.03948	39.999	40.88202

88	51.25027	58.12253	58.88291	77.19824	43.93757	40.07952	40.89961
89	51.23375	58.02866	58.92516	77.32817	43.81095	40.02212	40.86145
90	51.2268	58.10329	58.78906	77.33601	43.77673	40.01885	40.76001
91	51.28273	57.92925	58.8373	77.33745	43.74488	39.99986	40.89705
92	51.30353	57.88825	58.69074	77.30084	43.82011	39.98854	40.78019
93	51.34685	57.92164	58.91178	77.2766	43.78982	39.86203	40.88776
94	51.2933	57.84298	58.74662	77.11506	43.79344	39.98494	40.77821
95	51.12515	57.83442	58.86679	77.24507	43.73393	39.81563	40.81912
96	51.22664	57.83131	58.67286	77.04532	43.81578	39.84386	40.81795
97	51.27708	57.84547	58.74149	77.13279	43.60124	39.88064	40.83567
98	51.27233	57.86187	58.67474	77.01385	43.53326	39.73598	40.74616
99	51.13314	57.76471	58.51102	77.04546	43.76035	39.81075	40.83684
100	51.28656	57.83611	58.65034	77.13371	43.58532	39.75204	40.58357
101	51.2822	57.86822	58.55722	76.99279	43.55325	39.79509	40.64812
102	51.1528	57.52876	58.58897	76.98556	43.58296	39.67245	40.64595
103	51.35593	57.71754	58.53376	77.07317	43.62241	39.63942	40.65649
104	51.15573	57.71985	58.47741	76.92566	43.51165	39.6677	40.54374
105	51.2924	57.67062	58.53571	76.94817	43.48254	39.68003	40.627
106	51.15916	57.5864	58.52238	76.95088	43.50602	39.70058	40.5544
107	51.23796	57.72044	58.48192	76.83336	43.48192	39.51776	40.6249
108	51.11914	57.51362	58.40677	76.95968	43.53875	39.49531	40.61543
109	51.08954	57.69271	58.29718	76.87435	43.35084	39.64021	40.63041
110	51.08016	57.56303	58.35052	76.77375	43.29459	39.60644	40.54017
111	51.18895	57.58798	58.34612	76.82695	43.35902	39.49331	40.49708
112	51.41094	57.48739	58.27153	76.76607	43.35096	39.42682	40.52786
113	51.15869	57.58557	58.34897	76.71619	43.32811	39.46541	40.38505
114	51.22311	57.52237	58.19078	76.8028	43.33751	39.4448	40.39321
115	51.22559	57.47085	58.18827	76.83649	43.28847	39.43806	40.45873
116	51.135	57.35536	58.1796	76.67145	43.44002		40.50759
117	51.14015	57.44238	58.25297	76.68738	43.35246		40.56854
118	51.22696	57.43814	58.12094	76.63135	43.28229		40.47559
119	51.2296	57.33558	58.11858	76.5876	43.38659		40.49499
120	51.19598	57.46824	58.04436	76.65699	43.16662		40.25667
121	51.04011	57.32578	57.99168	76.63795	43.19512		40.25049
122	51.04963	57.42391	58.11069	76.56838	43.14332		40.37411
123	51.17508	57.29071	58.06806	76.43761	43.13617		40.42897
124	51.04264	57.30509	58.07108	76.46173	43.15598		40.29562
125	50.98176	57.13359	57.93523	76.43671	43.04309		40.40843
126	51.06868	57.29927	57.93825	76.40522	43.11992		40.38984
127	51.02869	57.20198	58.14308	76.57573	43.0497		40.30758
128	50.96969	57.18385	58.05438	76.46276	43.06785		40.27181
129	50.98327	57.13767	58.04407	76.45598	43.02283		40.34906
130	51.03134	56.97727	57.9824	76.4679	42.95269		40.28116
131	50.82867	57.13404	57.84585	76.44422	43.08034		40.42494

132	50.83406	57.27156	57.88649	76.30919	43.01569	40.21799
133	50.937	57.19021	57.93051	76.37545	42.89284	40.25388
134	50.87137	57.16351	57.7593	76.31072	42.89856	40.35557
135	50.85731	57.06792	57.87064	76.36718	42.88529	40.23458
136	50.90815	57.14652	57.69119	76.36141	42.88873	40.19958
137	50.74575	57.04083	57.64259	76.26462	42.94248	40.10696
138	50.84358	57.04918	57.64058	76.27164	42.86585	40.21517
139	50.76776	57.03993	57.7779	76.25264	42.88341	40.11473
140	50.76013	56.899	57.76524	76.13502	42.95658	40.13024
141	50.70208	57.10953	57.75009	76.19767		40.10312
142	50.67943	57.03345	57.77751	76.12801		40.17593
143	50.6487	56.91634	57.79796	76.19088		40.16965
144	50.56382	56.86415	57.67709	76.09403		40.162
145	50.63157	56.94647	57.73662	76.10984		40.29321
146	50.51439	56.94327	57.61863	76.0319		40.15692
147	50.41318	56.89829	57.53194	76.12576		40.05799
148	50.51145	56.83491	57.44199	76.01269		40.26886
149	50.65522	56.67815	57.72501	76.20361		40.13647
150	50.45054	56.84557	57.62684	76.03666		40.05505
151	50.45782	56.85045	57.11811	76.05113		40.06314
152	50.40738	56.78234	55.78127	76.06873		39.96376
153	50.566	56.68513	55.10135	75.9349		39.9612
154	50.50502	56.73582	56.23158	75.98221		40.1102
155	50.42616		57.04471	75.99089		40.02485
156	50.36898		57.36836	75.93994		40.0337
157	50.30708		57.57266	76.01919		39.99299
158	50.2776		57.58861	75.92619		39.90681
159	50.31396		57.65663	75.95962		39.87854
160	50.40371		57.66525	75.87479		39.96111
161	50.22034		57.80418	75.90004		39.88021
162	50.26112		57.82845	75.88817		40.03852
163	50.23983		57.72368	75.91265		39.9149
164	50.30138		57.83403	75.83961		39.85381
165	50.28145		57.98292	75.7604		39.84223
166	50.2721		57.81957	75.72317		39.87042
167	50.28104		57.81231	75.8051		39.7855
168	50.20655		57.78153	75.7406		39.84816
169	50.13794		57.79635	75.70303		39.83732
170	50.1557		57.83701	75.6517		39.91968
171	50.2235		57.72343			39.93974
172	50.2137		57.75319			39.65689
173	50.18541		57.67318			39.70156
174	50.14602		57.78529			39.7758
175	50.16425		57.74155			39.69199

176	50.10648	57.75815	39.8356
177	50.20001	57.74816	39.68075
178	50.17041	57.69889	39.61103
179	50.01204	57.83221	39.65002
180	50.12621	57.84627	39.58903
181	50.27059	57.78682	39.78267
182	50.00676	57.82139	39.55353
183	50.02595	57.77435	39.68023
184	49.95187	57.67262	39.5573
185	50.06773	57.82308	39.59505
186	50.04599	57.72171	39.63921
187	49.92996	57.59822	39.62605
188	49.98854	57.6846	39.49631
189	49.905	57.7796	39.49681
190	50.04343	57.58096	39.49972
191	49.99713	57.59556	39.52947
192	49.85089	57.67971	39.50394
193	49.79455	57.6035	39.45651
194	49.86569	57.60785	39.45141
195	49.91372	57.47566	39.48288
196	49.98011	57.39082	39.43947
197	49.71208	57.56828	39.50523
198	49.8606	57.53306	39.31
199	49.89504	57.54645	39.46431
200	49.82193	57.55546	39.45574
201	49.86911	57.42275	39.19507
202	49.71142	57.54615	39.42123
203	49.85111	57.4425	
204	49.72965	57.39623	
205	49.78557	57.27772	
206	49.79544	57.38273	
207	49.62812	57.2555	
208	49.79805	57.16659	
209	49.5818	57.28101	
210	49.62353	57.24488	
211	49.55992	57.14433	
212	49.58412	57.21225	
213	49.60812	57.17702	
214	49.58237	57.15863	
215	49.60215	57.18852	
216	49.47857	57.07025	
217	49.54528	57.17213	
218	49.45171	57.16507	
219	49.47642	57.07404	

220	49.58605	57.03399
221	49.58039	57.06201
222	49.66372	57.02143
223	49.57701	57.07863
224	49.44795	
225	49.45833	
226	49.62753	
227	49.29144	
228	49.52057	
229	49.45075	
230	49.47349	
231	49.45636	
232	49.4161	
233	49.48652	
234	49.42663	
235	49.33715	
236	49.42551	
237	49.38666	
238	49.34136	
239	49.20036	
240	49.26575	
241	49.22638	
242	49.21595	
243	49.26986	
244	49.14098	
245	49.20709	
246	49.09619	
247	49.12252	

50-0lb-one	50-0lb-no
50.51362	49.55157
50.51514	45.20186
50.27681	43.27471
50.3463	41.75646
50.36607	40.11266
50.40898	38.44718
50.38343	37.50948
50.22715	36.7788
50.33077	36.5536
50.17677	36.28608
50.34855	36.13822
50.18116	36.15903
50.15122	36.01338
49.99495	35.96805
50.14578	35.9299
50.22194	35.8604
50.07522	35.75524
50.15024	35.75662
49.98714	35.83845
50.02687	35.73153
49.89313	35.77232
50.01974	35.72582
49.91785	35.67983
49.89985	35.61508
49.88351	35.57194
49.80488	35.57951
49.63963	35.52989
49.73842	35.51462
49.66431	35.68442
49.7554	35.62019
49.60683	35.49176
49.70946	35.51748
49.54956	35.68416
49.72115	35.34325
49.5548	35.5697
49.55434	35.49418
49.46539	35.43346
49.50893	35.41395
49.38176	35.51608
49.54679	35.42532
49.25467	35.36363
49.35811	35.32863
49.30949	35.34042

49.29116	35.46933
49.10569	35.30245
49.39562	35.37334
49.33859	35.12276
49.24276	35.44392
49.13546	35.26641
49.18771	35.13497
49.17885	35.1674
49.22194	35.21636
49.13107	35.01193
49.16294	35.20714
49.1844	35.25649
49.22807	35.02791
49.11188	35.28037
49.20158	35.11715
49.16753	35.12033
49.06465	35.25291
49.18033	35.08925
49.18785	35.07736
49.14266	35.18413
49.14611	34.98419
49.07674	35.09228
48.991	35.05875
49.0288	34.91827
48.99897	34.95404
49.08192	35.16887
48.97415	34.91156
49.0063	35.03889
48.91746	34.84191
48.84486	34.82878
48.9281	34.77361
48.72223	34.8432
48.78343	34.6887
48.72319	34.73353
48.95266	34.71109
48.62504	34.7319
48.9005	34.77488
48.74078	34.57661
48.6615	34.60669
48.67364	34.6553
48.56639	34.71202
48.62928	34.64737
48.67621	34.6506
48.57276	34.51325

48.4559	34.76102
48.41158	34.77991
48.45998	34.73537
48.47145	34.49282
48.26944	34.6235
48.28959	34.68886
48.27263	34.61367
48.287	34.42491
48.22612	34.54185
48.23986	34.43427
48.24523	34.53965
48.21703	34.42383
48.22801	34.59472
48.17254	34.39959
47.99952	34.45533
48.22946	34.54133
48.03509	34.61289
48.11222	34.28396
48.15427	34.3854
47.86249	34.34758
48.04081	34.23242
48.0527	34.43652
47.96348	
47.7666	
47.90499	
47.93792	
47.76121	
47.98627	
47.86767	
47.83052	
47.82025	
47.80913	
47.6696	
47.74777	
47.65681	
47.84643	
47.72936	
47.59901	
47.77015	